EE6094
CAD for VLSI Design

PA1
Benchmark Translator

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Outline

- Input format
- Output format
- Evaluation

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Input format

- Your program takes the Netlist file (.netlist) as input

```
* Netlist of the module c17

Inst gat1 nand
Inst gat2 nand
Inst gat3 nand
Inst gat4 nand
Inst gat5 nand
Inst gat6 nand

NET net1 2
PIN inpt1, gat1/IN1
NET net2 2
PIN inpt2, gat3/IN1
NET net3 3
PIN inpt3, gat1/IN2, gat2/IN1
NET net6 2
PIN inpt6, gat2/IN2
NET net7 2
PIN inpt7, gat4/IN2
NET net10 2
PIN gat1/OUT1, gat5/IN1
NET net11 3
PIN gat2/OUT1, gat3/IN2, gat4/IN1
NET net16 3
PIN gat3/OUT1, gat5/IN2, gat6/IN1
NET net19 2
PIN gat4/OUT1, gat6/IN2
NET net22 2
PIN outpt22, gat5/OUT1
NET net23 2
PIN outpt23, gat6/OUT1
```

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Output format

- Your program will generate a synthesizable gate-level Verilog file (.v)

```
`timescale 1ns/1ps
module c17 (N1,N2,N3,N6,N7,N22,N23);

input N1,N2,N3,N6,N7;

output N22,N23;

wire N10,N11,N16,N19;

nand NAND2_1 (N10, N1, N3);
nand NAND2_2 (N11, N3, N6);
nand NAND2_3 (N16, N2, N11);
nand NAND2_4 (N19, N11, N7);
nand NAND2_5 (N22, N10, N16);
nand NAND2_6 (N23, N16, N19);

endmodule
```

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Evaluation

- Correctness (50%)
- Readability (10%)
- The report (10%)
- Demo session (30%)
- Bonus (10%)

Correctness

- We will use the Cadence NC-Verilog simulator to verify the correctness of your output
  - Generate an executable file
  - Source Cadence and Synopsys EDA tools
  - Use the executable file to translate the netlist file to Verilog file
  - Use the Cadence NC-Verilog simulator to verify the correctness

Readability

- Functions
- The name of the variables
- Comments
- Formatting

The report

- In your report, you have to include at least:
  - Compilation and execution
  - Completion
  - Hardness
  - Suggestions
- We don't restrict the report format and length

The report

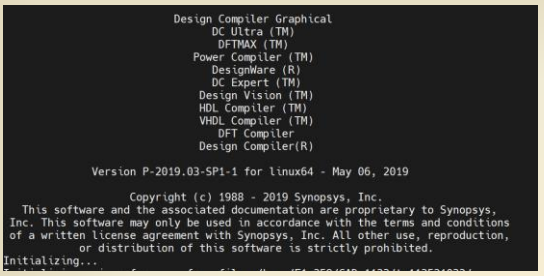
- You can also include
  - Cover
  - Flow charts
  - Code explanations
    - Concepts, functions, variables, etc.
- The grading of the report will compare yours with others
- Remind the resolution of the figures

Demo session

- You must show up in the demo session
- Be familiar with your code

Bonus

- Logic synthesis by Design Compiler



- Answer the questions in your report
- Also, submit all required output files
- Details would be released in another file

Q&A

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